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UG Program in Electronics and Computer Science

Honours in Emerging Area offered by Department of Artificial Intelligence and Data Science

Micro-Project Title

ClinXAI: Transparent Multi-Modal Clinical Diagnostics Using LIME and SHAP

Student Details:

Name: [STUDENT NAME]

PRN: [PRN]

Class and Division: [CLASS / DIVISION]

Academic Year: 2026-27

Subject: Trustworthy and Responsible AI (Course Code: **ADCOR1PC303**)

Guide: Mr. Milind S. Khairrnar

Submitted as partial fulfillment of the term-work for the subject,
Trustworthy and Responsible AI (Course Code: **ADCOR1PC303**)

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Certificate

This is to certify that the report of the project entitled

ClinXAI: Transparent Multi-Modal Clinical Diagnostics Using LIME and SHAP

is a bonafide work of

Name of student	Class	Roll No.
[STUDENT NAME]	T. Y.	[ROLL NUMBER]

submitted to the

Shah & Anchor Kutchhi Engineering College, Mumbai

during semester V in partial fulfilment of the requirement for the term-work of the subject, **"Trustworthy and Responsible AI" in Third Year, Semester V of B. Tech. Degree in**

Artificial Intelligence and Data Science

Guide

Mr. Milind S. Khairrnar

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Approval for Project Report for T. Y. Semester V

This project report entitled THIRD YEAR MICRO-PROJECT, "ClinXAI: Transparent Multi-Modal Clinical Diagnostics Using LIME and SHAP" by Student Name [STUDENT NAME] is approved for semester V in partial fulfilment of the requirement for the CIAP for the subject Trustworthy and Responsible AI (Course Code: **ADCOR1PC303**).

Examiner

1._____

Date:

Place:

UG Program in Electronics and Computer Science

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Name of student	Class	Roll No.	Signature
[STUDENT NAME]	T. Y.	[ROLL NUMBER]	

Date:

Place:

COURSE ARTICULATION & OUTCOME ALIGNMENT SHEET

Project Activity Component	Mapped CO	Bloom's Level	Mapped POs / PSOs	Measurable Performance Benchmark	ClinXAI Evidence
Explainability Architecture & Feature Attribution (SHAP / LIME)	CO 1	BL 2 (Understand) / BL 3 (Apply)	PO1, PO2, PO5, PSO1	Global summary & local force plots generated with zero syntax error	SHAP, LIME, waterfall and force plots; 24/24 tests
End-to-End Responsible AI Deployment & Interactive UI Dashboard	CO 1 - CO 3	BL 5 (Evaluate) / BL 6 (Create)	PO2, PO3, PO5, PO9, PSO3	Fully responsive live dashboard deployed on Streamlit or Gradio	Local 7-tab Streamlit app (not publicly deployed); 34/34 browser checks

Official rows unchanged; the other four components (fairness, adversarial, privacy, model card) are not implemented.

CIAP 25-MARK RUBRIC EVALUATION SHEET

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Definition & Scope (15%)	Clearly defined problem with a focused and achievable scope.	Problem identified, but scope may be unclear or too broad/narrow.	Problem vaguely defined, or scope missing.	Problem not clearly identified or missing.
Design & Methodology (20%)	Innovative and well-structured approach with justification for chosen methods.	Appropriate design and methodology, but limited justification.	Basic design and methodology, with some justification.	Weak or missing design and methodology.
Implementation & Results (25%)	Project completed effectively with clear and well-presented results.	Project completed, but results unclear or poorly presented.	Project partially completed with limited or unclear results.	Project not completed or results missing.
Analysis & Interpretation (20%)	Thorough analysis with insightful interpretation of results.	Analysis attempts were made, but interpretation could be improved.	Basic analysis with limited interpretation.	Weak or missing analysis and interpretation.
Report & Presentation (20%)	Clear, concise, and well-organized report with effective presentation.	Report generally well-written but may lack clarity or organization. Presentation could be improved.	Report lacks clarity or organization. Presentation needs significant work.	Poorly written report with ineffective or missing presentation.

Marks: _____ / 25 Evaluator's signature: _____ Date: _____

ABSTRACT

Accurate machine-learning models for breast-cancer diagnosis, such as Random Forests, act as black boxes. ClinXAI builds a transparent pipeline on the Breast Cancer Wisconsin (Diagnostic) dataset (569 patients, 30 nuclear features from fine-needle-aspirate images), with malignant as the positive class and a stratified 80/20 split keeping 114 patients unseen. A 300-tree Random Forest is compared with a Logistic Regression baseline, and every prediction is explained by LIME (a seeded local linear surrogate) and TreeSHAP (exact Shapley attributions in probability units), presented in a seven-tab Streamlit dashboard that supports patient selection, what-if biomarker editing and side-by-side comparison of the two explanations, with all results reproducible from fixed random seeds. On the held-out set the forest achieved 97.37% accuracy, 100% precision, 92.86% recall, 96.30% F1 and ROC-AUC 0.9944 (TN = 72, FP = 0, FN = 3, TP = 39); SHAP additivity held to 4.4×10^{-16}. LIME and SHAP shared on average 3.66 of their top-five features (Jaccard 0.62, Spearman 0.70), and 28.9% of patients showed a directional disagreement. Agreement was complete for confident predictions but collapsed for the missed malignancies (Patients #86 and #73), where LIME's local fit fell to $R^2 \leq 0.06$, showing that explanations must themselves be audited before clinical use.

Keywords: Explainable AI; LIME; SHAP; TreeSHAP; Random Forest; Breast Cancer Diagnosis; Clinical Decision Support

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LIST OF ABBREVIATIONS

Abbr.	Expansion	Abbr.	Expansion
AUC	Area Under the ROC Curve	LR	Logistic Regression
CO / PO	Course / Program Outcome	RF	Random Forest
FN / FP	False Negative / Positive	ROC	Receiver Operating Characteristic
FNA	Fine-Needle Aspirate	SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations	XAI	Explainable Artificial Intelligence

CHAPTER 1: PROBLEM DEFINITION & OPERATIONAL RELEVANCE

1.1 Problem Statement and Background

Breast cancer is the most common cancer in women (about 2.3 million diagnoses and 670,000 deaths in 2022 [1]). Nuclear measurements from digitised fine-needle-aspirate images [2] form the Breast Cancer Wisconsin (Diagnostic) dataset [3]. Accurate ensembles such as a 300-tree Random Forest (10,976 nodes here) are black boxes [4], [5]. **Problem:** predict P(malignant) from 30 features, explain each prediction with LIME and SHAP for the same class, and measure when the two explanations disagree.

1.2 Clinical and Societal Relevance

Unexplained scores invite blind trust or rejection, a missed cancer with a plausible explanation is hard to catch, and explainers can disagree [6]; regulators expect transparency and human oversight [7], [8].

1.3 Project Objectives

O1 Implement and evaluate a Random Forest against a Logistic Regression baseline on 114 held-out patients. **O2** Apply LIME and TreeSHAP and verify SHAP additivity ($< 10^{-6}$). **O3** Quantify LIME–SHAP agreement (top-5 overlap, Jaccard, Spearman ρ, sign agreement, seed stability). **O4** Design and validate a seven-tab dashboard.

1.4 Project Scope and Operational Boundaries

In scope: tabular prediction, LIME/SHAP explanation and comparison, five case studies, a local dashboard. Out of scope: raw images, calibrated risk, treatment advice, clinical validation, fairness audit, public deployment. *Multi-modal* refers to the two explanation modalities; the input is one tabular modality derived from images.

CHAPTER 2: THEORETICAL FORMULATION, LITERATURE REVIEW & MATHEMATICAL FRAMEWORK

2.1 Explainable Artificial Intelligence

XAI explanations are post-hoc or intrinsic, local or global, and agnostic or model-specific; this report evaluates **local fidelity**, **consistency**, **stability** and computational cost.

2.2 Random Forest Classification

$$f(x) = (1/T) \cdot \Sigma_{t=1..T}\, p_t(y = 1 \mid x), \qquad G(node) = 1 - \Sigma_k\, p_k^2 \tag{2.1}$$

The forest averages T = 300 bootstrap trees [4] grown with Gini impurity G and $\lfloor\sqrt{30}\rfloor = 5$ features per split (scikit-learn); p_t is the malignant leaf fraction of tree t.

2.3 LIME: Local Interpretable Model-Agnostic Explanations

LIME [9] fits a simple model g that imitates f only near x, using quartile-binned rules such as `worst area > 1122.50` → +0.137.

2.4 LIME Perturbation and Local Surrogate Formulation

$$\xi(x) = \operatorname{argmin}_g\, \Sigma_i\, \pi_x(z_i)\cdot(f(z_i) - g(z'_i))^2 + \lambda\|w\|^2 \tag{2.2}$$

Here z_i are 5,000 perturbations, $z' \in \{0,1\}^{30}$ marks whether each feature stays in the patient's bin, $\pi_x = \exp(-D^2/2\sigma^2)$ with $\sigma = 0.75\sqrt{30} \approx 4.108$, $\lambda = 1$ and $g(z') = w_0 + \Sigma\, w_j z'_j$. The weighted R^2 is LIME's only fidelity signal, and results depend on the seed [10].

2.5 SHAP: Shapley Additive Explanations

$$P(\text{malignant} \mid x) = \phi_0 + \Sigma_{j=1..30}\, \phi_j(x), \qquad \phi_0 = E[f(X)] \tag{2.3}$$

SHAP [11] attributions satisfy local accuracy (Equation 2.3), missingness and consistency.

2.6 Shapley Game-Theoretic Attribution

$$\phi_j(x) = \Sigma_{S\subseteq F\setminus\{j\}}\, [|S|!\cdot(M-|S|-1)!/M!]\cdot[v_x(S\cup\{j\}) - v_x(S)] \tag{2.4}$$

Features are players and $v_x(S) = E[f(X) \mid X_S = x_S]$ is a coalition's payout [12]; naive evaluation needs $2^{30} \approx 1.07 \times 10^9$ coalitions.

2.7 TreeSHAP Explanation Method

TreeSHAP [13] computes Equation (2.4) exactly in $O(T \cdot L \cdot D^2)$ using each node's training cover; $\varphi_0 = 0.3747$ ($\approx$ training prevalence 170/455) and the φ_j share probability units with `predict_proba`.

2.8 LIME versus SHAP: Mathematical and Practical Trade-offs

LIME is approximate, seed-dependent and compares a bin with other bins; TreeSHAP is exact, deterministic and compares with E[f(X)]. Magnitudes are not comparable, so agreement uses ranks and signs of the top-5 sets A, B:

$$J = |A \cap B| / |A \cup B|, \qquad \rho = 1 - 6\Sigma_j d_j^2 / (n(n^2 - 1)), \quad n = 30 \qquad (2.5)$$

2.9 Comparative Literature Review

Logistic regression is faithful but restricted [5]; LIME and KernelSHAP are model-agnostic but approximate; TreeSHAP is exact for trees. Perturbation explainers can be fooled [14] and often disagree [6], which matters most in high-stakes medicine [5].

CHAPTER 3: SYSTEM ARCHITECTURE, DESIGN PIPELINE & WORKFLOW MODELS

3.1 Overall System Architecture

Figure 3.1 shows the data, model, explanation, comparison, presentation and test layers, each a real module (`src/`, `train_model.py`, `evaluate.py`, `app.py`, `tests/`).

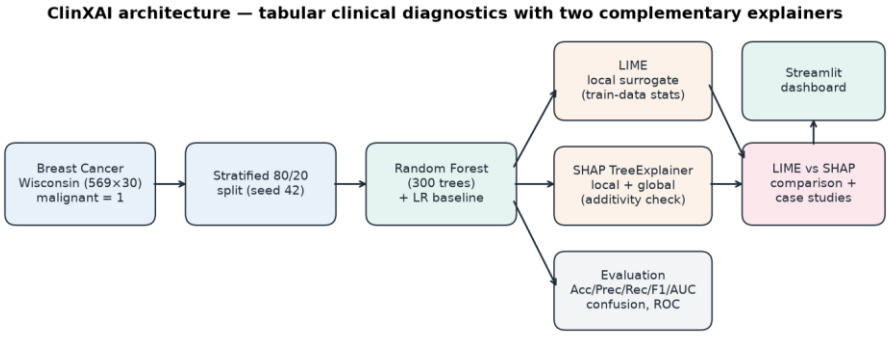


Figure 3.1: ClinXAI system architecture

3.2 Dataset and Feature Description

569 patients (212 malignant, 357 benign) [3]; ten nuclear characteristics (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) as mean, standard error and worst values = 30 features, no missing values.

3.3 Data Preprocessing Pipeline

Validation of shape and values; malignant re-encoded to 1 (asserted); stratified 80/20 split, random_state = 42 (455 / 114); model, LIME statistics and scaler fitted on training data only; feature order enforced.

3.4 Random Forest Diagnostic Prediction Pipeline

`train_model.py` trains the forest, asserts `classes_ == [0, 1]`, evaluates at threshold 0.5, runs 5-fold CV and saves one joblib bundle.

3.5 LIME Explanation Workflow

Fresh seeded explainer → 5,000 perturbations → forest via a feature-name wrapper → kernel weights → ridge surrogate (weights, rules, R^2).

3.6 SHAP Explanation Workflow

`TreeExplainer` → (n, 30, 2) array → malignant slice → additivity check → global, waterfall and force plots.

3.7 Streamlit User Interaction Flow

Sidebar patient choice and what-if edits → `validate_patient_input()` → cached LIME/SHAP → seven tabs.

3.8 System Architecture Diagram and Data Flow Diagram

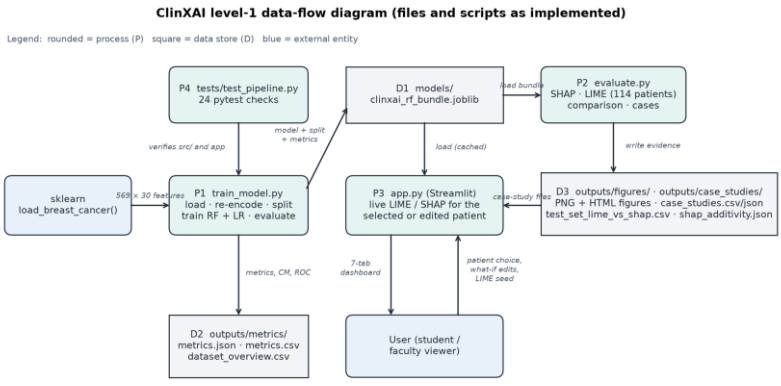


Figure 3.2: Level-1 data-flow diagram

CHAPTER 4: IMPLEMENTATION DETAILS, ALGORITHMS & TECHNOLOGICAL ARTIFACTS

4.1 Development Environment and Technology Stack

Python 3.12.10 on Windows 11 (CPU only); scikit-learn 1.9.1, LIME 0.2.0.1, SHAP 0.54.1, Streamlit 1.66.1, Matplotlib 3.11.2, pandas 3.1.0, NumPy 2.5.4, pytest 9.1.1 (pinned in `requirements.txt`).

4.2 Dataset Loading and Target Mapping

`load_dataset()` derives the mapping from `target_names`, re-encodes malignant = 1 and asserts it against the raw labels.

4.3 Model Training Implementation

`train_random_forest` fits `RandomForestClassifier(**RF_PARAMS)` on the training set and asserts `classes_ == [0, 1]`, so column 1 of `predict_proba` is always P(malignant).

4.4 LIME Explainer Implementation

A wrapper rebuilds a DataFrame in training order for every LIME query, and a fresh seeded explainer per call makes explanations reproducible; label 1 (malignant) and all 30 features are explained.

4.5 SHAP Explainer Implementation

`_select_malignant` normalises version-dependent SHAP outputs to the malignant class; `additivity_error` checks Equation (2.3) against `predict_proba`.

4.6 LIME–SHAP Comparison Implementation

`compare_explanations` computes top-5 sets, Jaccard, Spearman ρ and the sign table; `select_case_studies` picks max P, min P, P nearest 0.5, lowest Jaccard and most sign conflicts, reporting any missing case as not found.

4.7 Streamlit Interface Implementation

`app.py` (488 lines): sidebar selector and LIME settings, seven tabs, what-if editor, stability toggle and cached explanations (Figure 5.6).

4.8 Input Validation and Error Handling

`validate_patient_input` rejects missing features, NaN, ∞, text and negatives (7 tests); out-of-range values raise a warning.

4.9 Code Structure and Module Responsibilities

1,916 lines across `src/` (7 modules), three scripts and a 24-test suite (Appendix A).

CHAPTER 5: EXPERIMENTAL EVALUATION, TESTING, TRADE-OFF ANALYSIS & EMPIRICAL RESULTS

5.1 Experimental Methodology

Fit on 455 patients, test once on 114 (threshold 0.5); LIME (seed 42) and TreeSHAP explain every test patient; five rule-selected case studies.

5.2 Dataset Characteristics and Train/Test Split

Train 455 (170 malignant); test 114 (42 malignant, 72 benign).

5.3 Accuracy, Precision, Recall, F1 and ROC-AUC

Table 5.1: Held-out test performance (positive class = malignant)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	97.37%	100.00%	92.86%	96.30%	0.9944
Logistic Regression	96.49%	97.50%	92.86%	95.12%	0.9960

5.4 Confusion Matrix and ROC Curve

TN = 72, FP = 0, FN = 3, TP = 39: every error is a missed malignancy (Figure 5.1). CV ROC-AUC on training data: 0.9880 ± 0.0073.

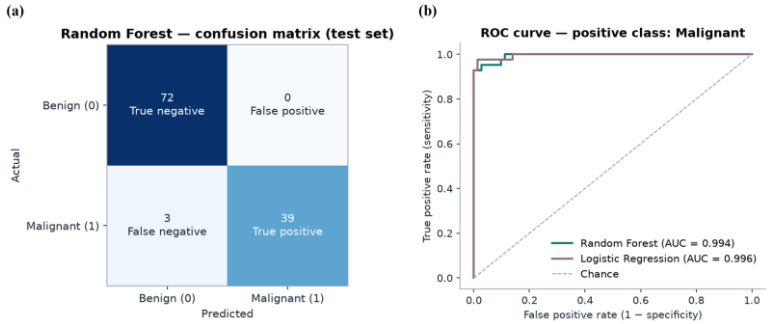


Figure 5.1: (a) Confusion matrix; (b) ROC curves

5.5 Global SHAP Feature Importance

Top features by mean |SHAP|: worst area (0.067), worst perimeter (0.060), worst concave points (0.059), mean concave points (0.044) (Figure 5.2).

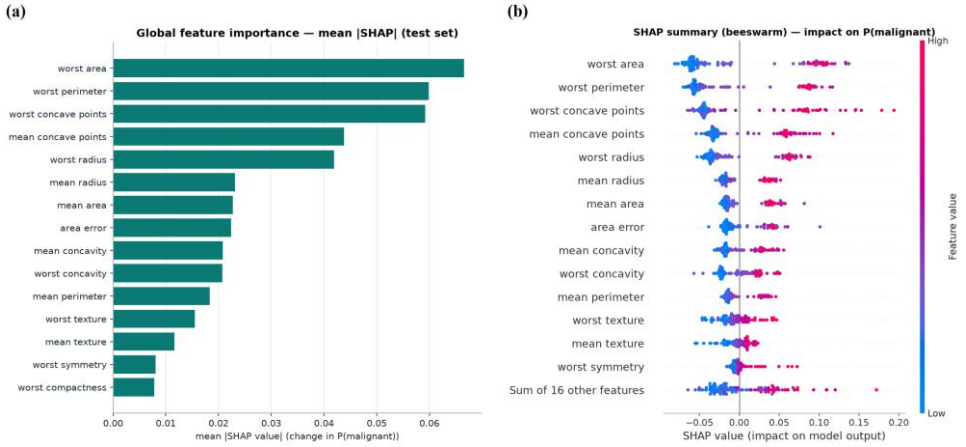


Figure 5.2: (a) Global mean |SHAP|; (b) SHAP beeswarm

5.6 Individual Patient LIME Explanations

For #70 (P = 1.000) every top LIME rule points to malignant (R^2 = 0.63). For missed malignancy #86 (P = 0.297) the top rules point to malignant despite a benign output and the surrogate barely fits (R^2 = 0.03; predicts 0.489) (Figure 5.3a).

5.7 Individual Patient SHAP Explanations

For #86, 0.3747 − 0.0780 = 0.2967 (Figure 5.3b). Additivity held for all 114 patients (max error **4.4e-16**).

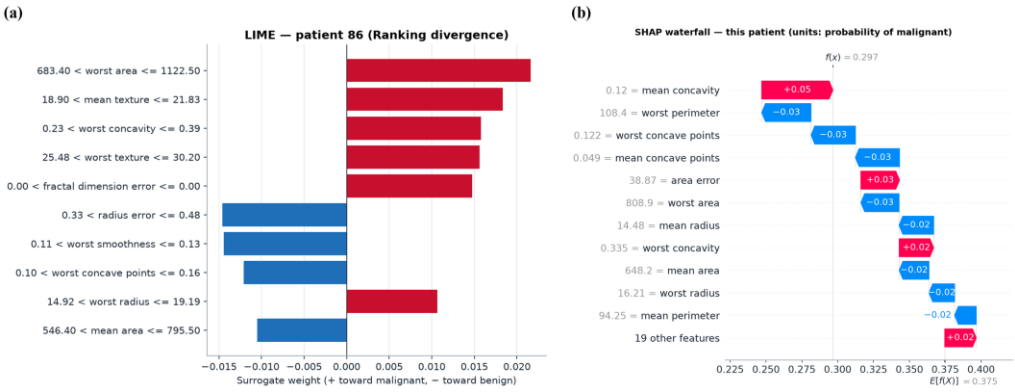


Figure 5.3: Patient #86: (a) LIME and (b) SHAP waterfall

5.8 LIME–SHAP Case Study Comparison

Table 5.2: Five programmatically selected case studies

Case	Patient	P(mal.)	Actual / predicted	Top-5	Jaccard	ρ	Sign flips	R^2
Clearly malignant	#70	1.000	Malignant / Malignant	5/5	1.00	0.90	0	0.63
Clearly benign	#74	0.000	Benign / Benign	5/5	1.00	0.83	0	0.17
Borderline	#100	0.513	Malignant / Malignant	2/5	0.25	0.60	2	0.05
Ranking divergence	#86	0.297	Malignant / Benign	0/5	0.00	0.27	4	0.03
Direction disagreement	#73	0.160	Malignant / Benign	1/5	0.11	0.39	4	0.06

Confident cases #70 and #74 share all five features with no sign conflicts, although #74's surrogate is off by 0.11 (R^2 0.17). Borderline #100 shares only two. Missed malignancies #86 (**no shared feature**, seed stability 0.39) and #73 (one shared) each show four sign flips (Figure 5.4). Worst area for #86/#73 (808.9/812.4) lies in the wide bin 683.4–1122.5: LIME compares it with mostly benign lower bins (+), SHAP with the base value (−). **Clinical limitation:** SHAP gives an exact rationale for a wrong prediction and LIME gives malignant reasons for a benign label; neither is evidence about the tumour.

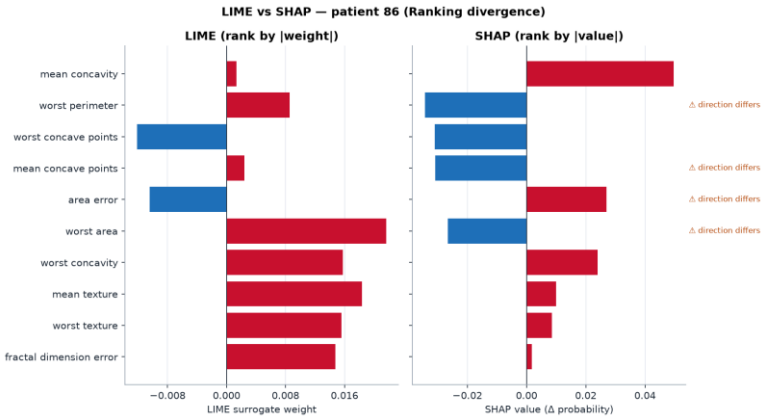


Figure 5.4: LIME vs SHAP for Patient #86

5.9 Explanation Divergence Analysis

Over 114 patients: mean top-5 overlap **3.66/5**, Jaccard **0.62**, Spearman **0.70**, direction disagreement in **28.9%** (33 patients) and mean LIME R^2 0.27. Agreement rises with confidence $|P - 0.5|$ ($\rho = 0.60$; Figure 5.5), and all three false negatives (#73, #86, #385) have Jaccard ≤ 0.25 and $R^2 \leq 0.061$ (descriptive analysis of the saved CSV). Causes: different reference points, a linear surrogate on a non-linear boundary, 21 feature pairs

with |r| > 0.9 and perturbation noise [14], [10]. **Low LIME fidelity and disagreement concentrate on the predictions that most need human review.**

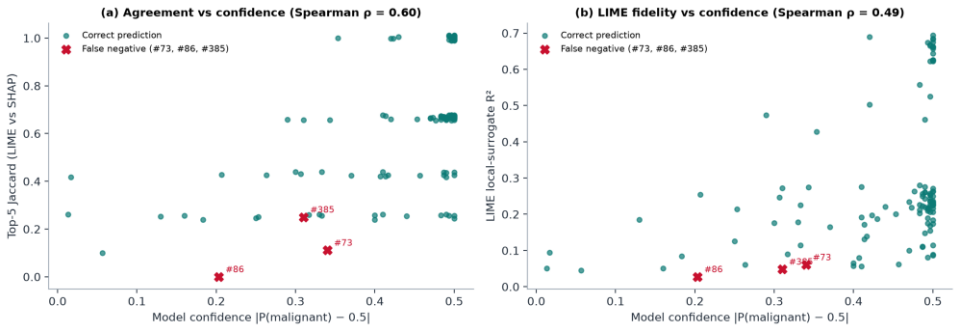


Figure 5.5: Agreement and LIME fidelity versus model confidence

5.10 Model Performance and Interpretability Trade-offs

The forest beats the transparent baseline by one false positive and both miss three cancers [5]; LIME trades fidelity for simplicity, TreeSHAP is exact but tree-only, and a lower threshold would need separate validation data.

5.11 Operational Verification Screenshots

Real screenshots (Edge, Playwright). Verification: 24/24 tests (also after a clean install), 35/35 output checks, 34/34 browser checks.

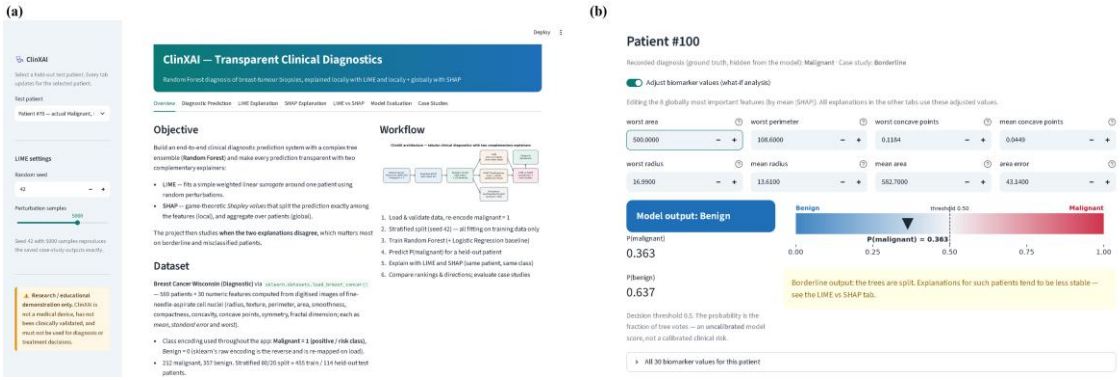


Figure 5.6: (a) Dashboard; (b) what-if on #100 flips the output to benign



Figure 5.7: Error handling: out-of-range input warning

CHAPTER 6: SOCIETAL IMPACT, ETHICAL GOVERNANCE, SAFETY & SECURITY AUDITS

6.1 Importance of Transparency in Clinical AI

Transparency lets clinicians check predictions [8]; ClinXAI also audits the explanations.

6.2 Human Oversight and Clinical Decision Support

ClinXAI is not a medical device and is not clinically validated: an educational aid with a disclaimer, an *uncalibrated* label, borderline/low-R^2/extrapolation warnings and no recommendations.

6.3 Patient Privacy and Responsible Data Usage

Public de-identified data only; real patient data would require consent and safeguards [15].

6.4 Model Bias, Dataset Limitations and Generalisation

Single-centre data, no demographics (no fairness audit), 114 test patients, possible distribution shift.

6.5 Safety Considerations and False Negatives

Three missed malignancies (P 0.160–0.297) with unreliable explanations; explanation manipulation is untested and the pickle-based model file has no checksum.

6.6 Relevant Ethical and Regulatory Considerations

No compliance, certification or clinical validation is claimed. The EU AI Act requires transparency, oversight and robustness for high-risk medical AI [7], and WHO guidance [8] requires validity, including clinical validation; India's ICMR (2023) AI guidelines stress the same.

6.7 Alignment with UN Sustainable Development Goal 3

Supports SDG 3.4 (reducing cancer mortality) [16] through a reproducible, CPU-only audit method.

CHAPTER 7: CONCLUSION, ENGINEERING LIMITATIONS & FUTURE ENHANCEMENTS

7.1 Summary of Work Completed

A tested pipeline: Random Forest, LIME and TreeSHAP for the same class, LIME–SHAP comparison, five case studies and a seven-tab dashboard.

7.2 Achievement of Project Objectives

O1 achieved (accuracy 97.37%, AUC 0.9944); O2 achieved (additivity 4.4×10^{-16}); O3 achieved (Jaccard 0.62, ρ 0.70); O4 achieved locally (34/34 browser checks).

7.3 Technical Limitations

Seed-dependent, low-fidelity LIME; path-dependent SHAP without a background set; small single-centre data; uncalibrated probabilities; no external or clinical validation; tabular input only.

7.4 Future Enhancements

Image + tabular multimodal models, LIME fidelity thresholds, interventional SHAP, probability calibration, external validation and **medical-expert validation before any deployment**.

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APPENDIX A: COMPLETE SOURCE CODE LISTINGS / CORE CONFIGURATION FILES

Condensed for length: the file inventory below points to the complete source code submitted in the project folder `ClinXAI/`; the key comparison routine follows.

Table A.1: Source-file inventory (complete code in the project folder)

File	Purpose	Lines	File	Purpose	Lines
app.py	Streamlit dashboard (7 tabs)	488	src/explain_lime.py	LIME explanations	94
train_model.py	Training, metrics, model bundle	89	src/explain_shap.py	SHAP explanations	80
evaluate.py	SHAP, LIME, comparison, case studies	183	src/comparison.py	LIME–SHAP agreement	125
src/data_loader.py	Loading, mapping, validation	124	src/visualization.py	Figures	298
src/model.py	Model training and evaluation	109	tests/test_pipeline.py	24 automated tests	282

Code excerpt A.1 — `src/comparison.py`, lines 29–37: LIME–SHAP agreement measures

```
def compare_explanations(lime_weights: pd.Series, shap_values: pd.Series, k: int =
    TOP_K) -> dict:
    """Compare two attribution vectors indexed by the same feature names (same
    patient, class)."""
    lime_weights = lime_weights.reindex(shap_values.index).fillna(0.0)
    lime_top, shap_top = top_k(lime_weights, k), top_k(shap_values, k)
    shared = [f for f in lime_top if f in shap_top]
    union = list(dict.fromkeys(lime_top + shap_top))

    rho, _ = spearmanr(lime_weights.abs().to_numpy(), shap_values.abs().to_numpy())
```

APPENDIX B: RAW DIAGNOSTIC TRACE LOGS, AUDIT TABLES & VERIFICATION SHEETS

`python -m pytest -v` — final lines of the 24-test log (report/evidence/pytest_verbose.txt)

```
tests/test_pipeline.py::test_invalid_numeric_input[<lambda>-Unknown] PASSED [ 91%]
tests/test_pipeline.py::test_invalid_input_length PASSED                 [ 95%]
tests/test_pipeline.py::test_streamlit_app_runs_and_updates_on_patient_change PASSED
    [100%]
============================== 24 passed in 17.82s ==============================
```

Table B.1: Verification summary

Verification layer	Result
Automated tests (also after a clean install)	24 / 24 passed
Independent saved-output validation	35 / 35 passed
Real-browser click-through (Microsoft Edge)	34 / 34 passed
Runtime warnings during training and evaluation	0